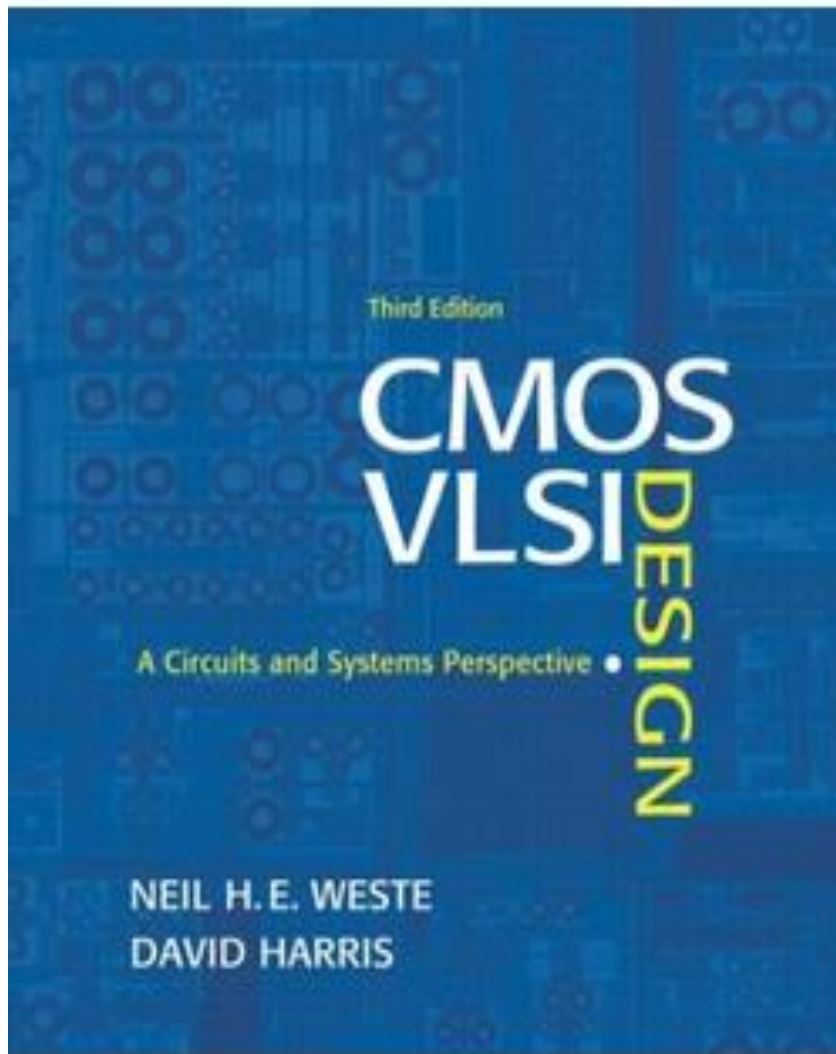




# **Lecture 4: Nonideal Transistor Theory**

# Outline

- ❑ Nonideal Transistor Behavior
  - High Field Effects
    - Mobility Degradation
    - Velocity Saturation
  - Channel Length Modulation
  - Threshold Voltage Effects
    - Body Effect
    - Drain-Induced Barrier Lowering
    - Short Channel Effect
  - Leakage
    - Subthreshold Leakage
    - Gate Leakage
    - Junction Leakage
- ❑ Process and Environmental Variations

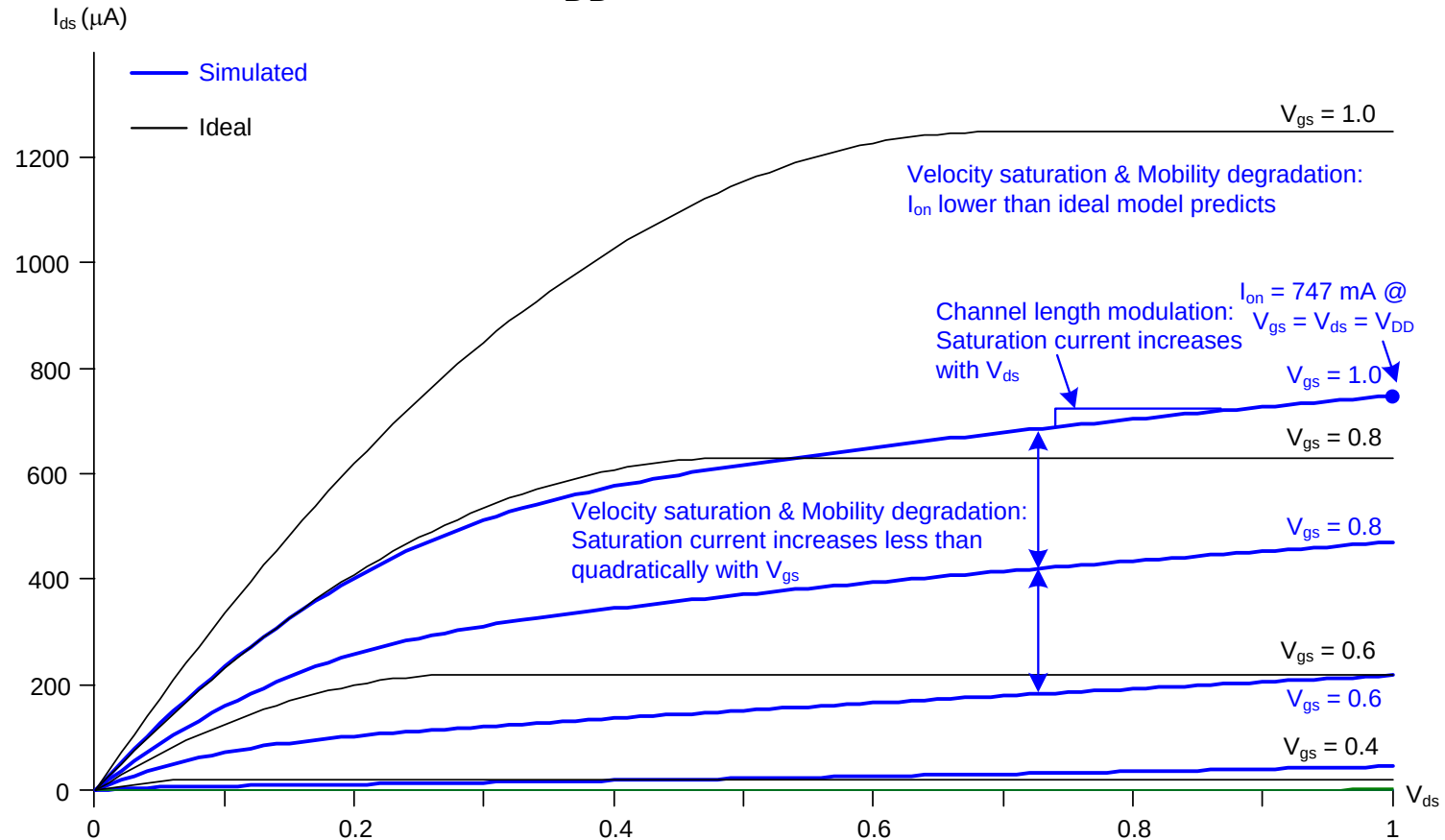
# Ideal Transistor I-V

❑ Shockley long-channel transistor models

$$I_{ds} = \begin{cases} 0 & V_{gs} < V_t & \text{cutoff} \\ \beta \left( V_{gs} - V_t - \frac{V_{ds}}{2} \right) V_{ds} & V_{ds} < V_{dsat} & \text{linear} \\ \frac{\beta}{2} (V_{gs} - V_t)^2 & V_{ds} > V_{dsat} & \text{saturation} \end{cases}$$

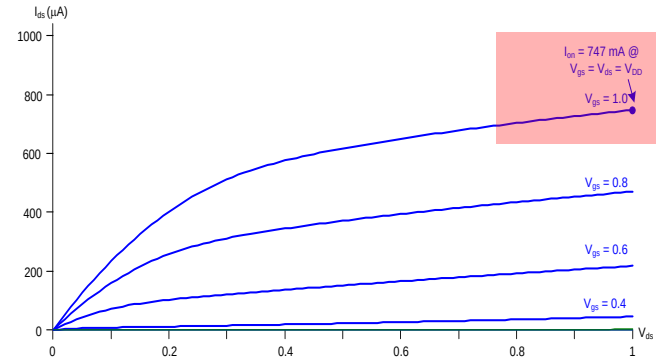
# Ideal vs. Simulated nMOS I-V Plot

□ 65 nm IBM process,  $V_{DD} = 1.0$  V

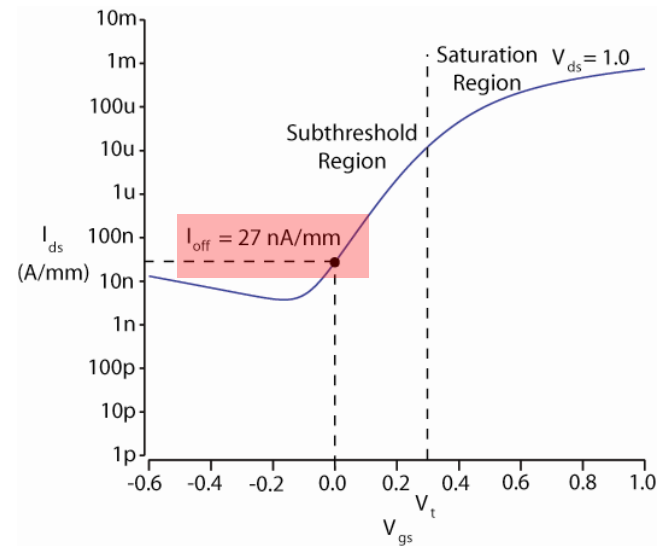


# ON and OFF Current

□  $I_{on} = I_{ds} @ V_{gs} = V_{ds} = V_{DD}$   
– Saturation



□  $I_{off} = I_{ds} @ V_{gs} = 0, V_{ds} = V_{DD}$   
– Cutoff



# Electric Fields Effects

- ❑ Vertical electric field:  $E_{\text{vert}} =$  \_\_\_\_\_
  - Attracts carriers into channel
  - Long channel:  $Q_{\text{channel}} \propto E_{\text{vert}}$
- ❑ Lateral electric field:  $E_{\text{lat}} =$  \_\_\_\_\_
  - Accelerates carriers from drain to source
  - Long channel:  $v = \mu E_{\text{lat}}$

# Mobility Degradation

- ❑ High  $E_{\text{vert}}$  effectively reduces mobility
  - Collisions with oxide interface

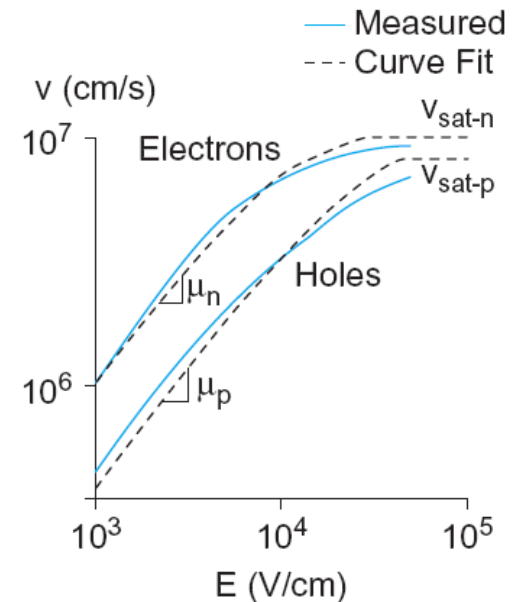
$$\mu_{\text{eff}-n} = \frac{540 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}}{1 + \left( \frac{V_{gs} + V_t}{0.54 \frac{\text{V}}{\text{nm}} t_{\text{ox}}} \right)^{1.85}}$$

$$\mu_{\text{eff}-p} = \frac{185 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}}{1 + \frac{|V_{gs} + 1.5V_t|}{0.338 \frac{\text{V}}{\text{nm}} t_{\text{ox}}}}$$

# Velocity Saturation

- ❑ At high  $E_{\text{lat}}$ , carrier velocity rolls off
  - Carriers scatter off atoms in silicon lattice
  - Velocity reaches  $v_{\text{sat}}$ 
    - Electrons:  $10^7$  cm/s
    - Holes:  $8 \times 10^6$  cm/s
  - Better model

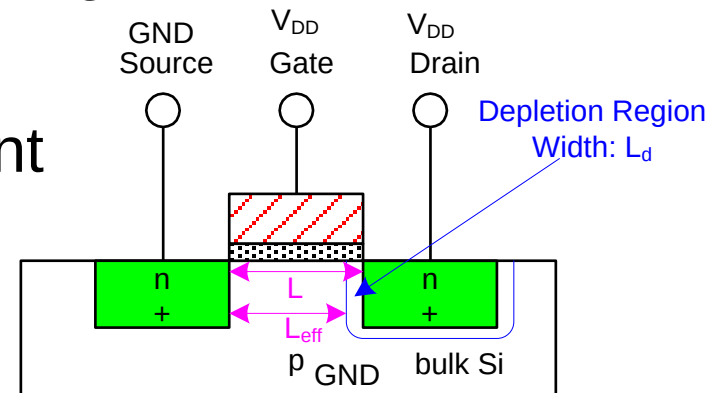
$$v = \begin{cases} \frac{\mu_{\text{eff}} E}{1 + \frac{E}{E_c}} & E < E_c \\ v_{\text{sat}} & E \geq E_c \end{cases} \quad E_c = \frac{2v_{\text{sat}}}{\mu_{\text{eff}}}$$





# Channel Length Modulation

- ❑ Reverse-biased p-n junctions form a *depletion region*
  - Region between n and p with no carriers
  - Width of depletion  $L_d$  region grows with reverse bias
  - $L_{\text{eff}} = L - L_d$
- ❑ Shorter  $L_{\text{eff}}$  gives \_\_\_\_\_ current
  - $I_{\text{ds}}$  \_\_\_\_\_ with  $V_{\text{ds}}$
  - Even in saturation



# Chan Length Mod I-V

$$I_{ds} = \frac{\beta}{2} (V_{gs} - V_t)^2 (1 + \lambda V_{ds})$$

- $\lambda$  = *channel length modulation coefficient*
  - not feature size
  - Empirically fit to I-V characteristics

# Threshold Voltage Effects

- ❑  $V_t$  is  $V_{gs}$  for which the channel starts to invert
- ❑ Ideal models assumed  $V_t$  is constant
- ❑ Really depends (weakly) on almost everything else:
  - Body voltage: *Body Effect*
  - Drain voltage: *Drain-Induced Barrier Lowering*
  - Channel length: *Short Channel Effect*

# Body Effect

- ❑ Body is a fourth transistor terminal
- ❑  $V_{sb}$  affects the charge required to invert the channel
  - Increasing  $V_s$  or decreasing  $V_b$  increases  $V_t$

$$V_t = V_{t0} + \gamma \left( \sqrt{\phi_s + V_{sb}} - \sqrt{\phi_s} \right)$$

- ❑  $\phi_s = \text{surface potential at threshold}$

$$\phi_s = 2v_T \ln \frac{N_A}{n_i}$$

- Depends on doping level  $N_A$
- And intrinsic carrier concentration  $n_i$

- ❑  $\gamma = \text{body effect coefficient}$

$$\gamma = \frac{t_{ox}}{\epsilon_{ox}} \sqrt{2q\epsilon_{si}N_A} = \frac{\sqrt{2q\epsilon_{si}N_A}}{C_{ox}}$$

# Body Effect Cont.

- For small source-to-body voltage, treat as linear

$$V_t = V_{t0} + k_\gamma V_{sb}$$

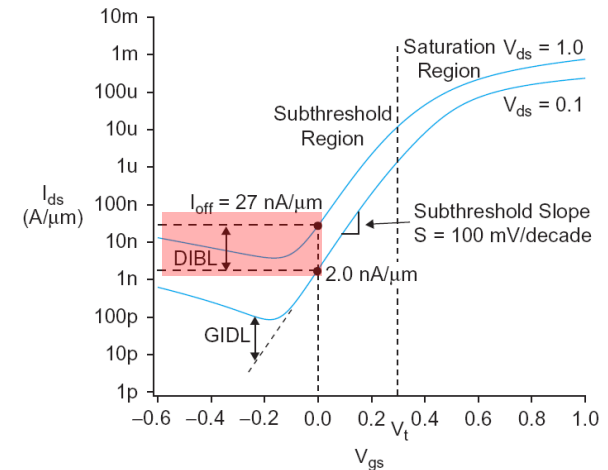
$$k_\gamma = \frac{\gamma}{2\sqrt{\phi_s}} = \frac{\sqrt{\frac{q\epsilon_{si}N_A}{v_T \ln \frac{N_A}{n_i}}}}{2C_{ox}}$$

# DIBL

- ❑ Electric field from drain affects channel
- ❑ More pronounced in small transistors where the drain is closer to the channel
- ❑ Drain-Induced Barrier Lowering
  - Drain voltage also affect  $V_t$

$$V_t' = V_t - \eta V_{ds}$$

- ❑ High drain voltage causes current to \_\_\_\_\_



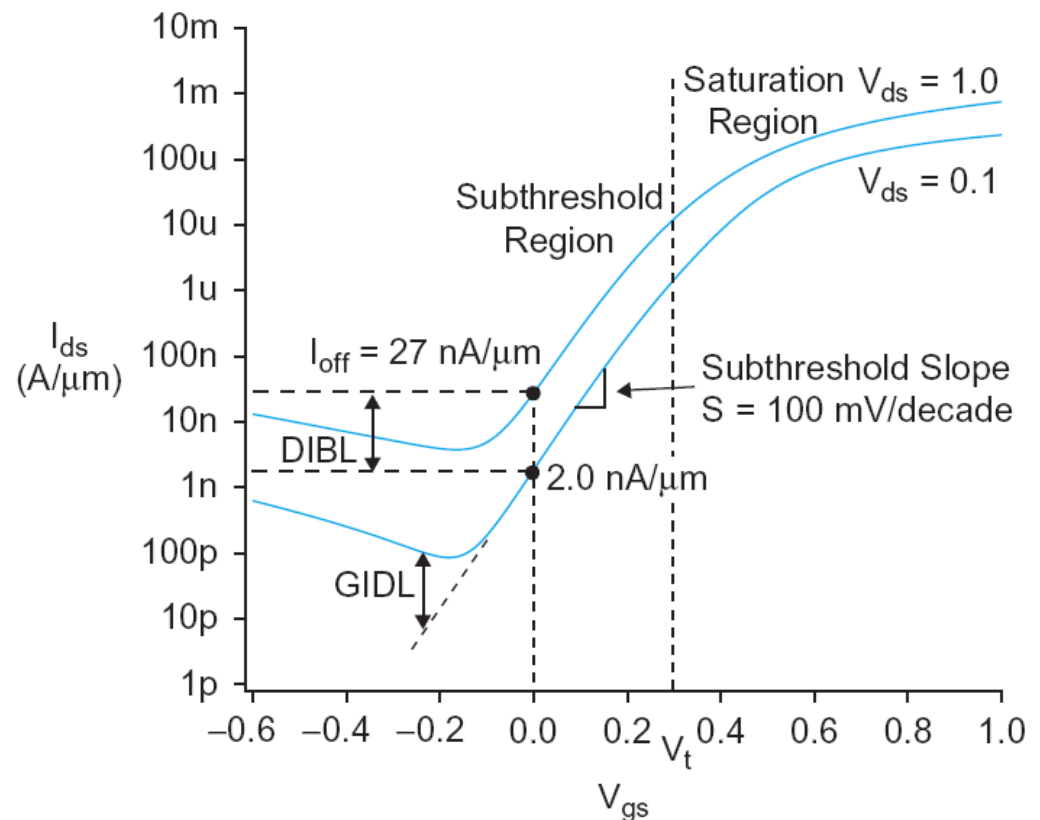
# Short Channel Effect

- ❑ In small transistors, source/drain depletion regions extend into the channel
  - Impacts the amount of charge required to invert the channel
  - And thus makes  $V_t$  a function of channel length
- ❑ Short channel effect:  $V_t$  increases with  $L$ 
  - Some processes exhibit a reverse short channel effect in which  $V_t$  decreases with  $L$

# Leakage

- ❑ What about current in cutoff?
- ❑ Simulated results
- ❑ What differs?

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# Leakage Sources

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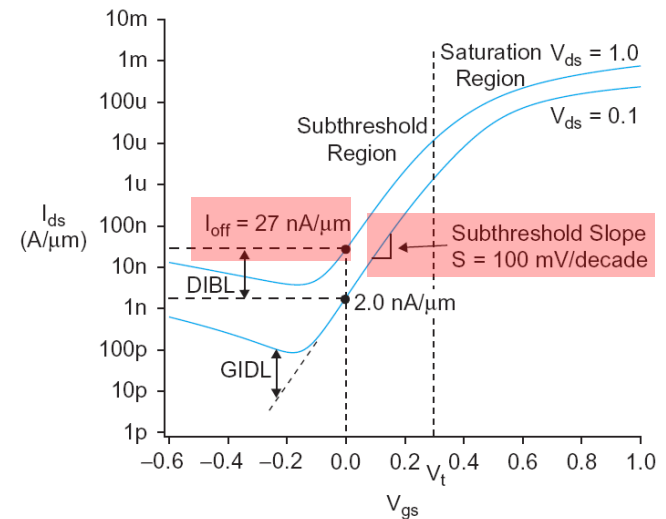
- ❑ Subthreshold conduction
  - Transistors can't abruptly turn ON or OFF
- ❑ Gate leakage
  - Tunneling through ultrathin gate dielectric
- ❑ Junction leakage
  - Reverse-biased PN junction diode current

# Subthreshold Leakage

- Subthreshold leakage exponential with  $V_{gs}$

$$I_{ds} = I_{ds0} e^{\frac{V_{gs} - V_{t0} + \eta V_{ds} - k_{\gamma} V_{sb}}{n v_T}} \left( 1 - e^{\frac{-V_{ds}}{v_T}} \right)$$

- $n$  is process dependent
  - typically 1.3-1.7



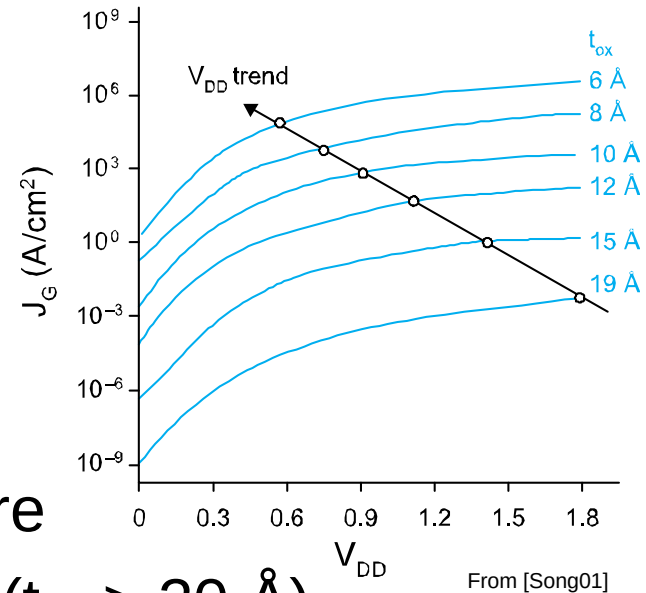
# Gate Leakage

- ❑ Carriers tunnel thorough very thin gate oxides
- ❑ Exponentially sensitive to  $t_{\text{ox}}$  and  $V_{\text{DD}}$

$$I_{\text{gate}} = WA \left( \frac{V_{\text{DD}}}{t_{\text{ox}}} \right)^2 e^{-B \frac{t_{\text{ox}}}{V_{\text{DD}}}}$$

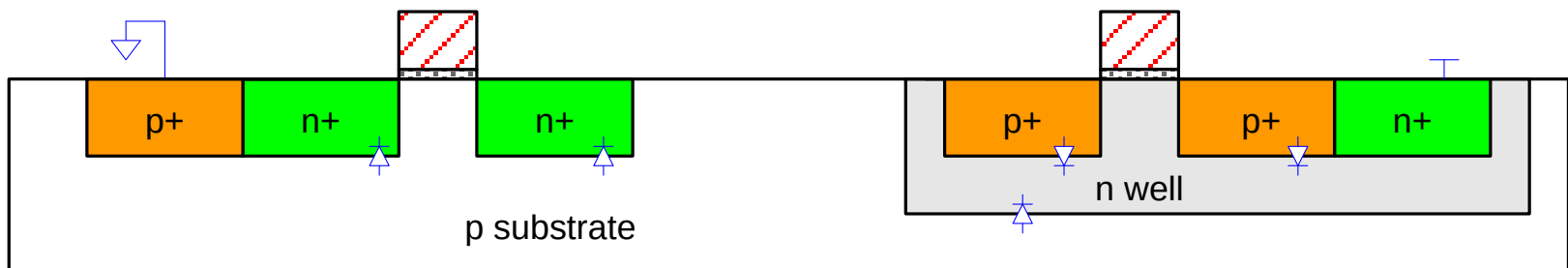
- A and B are tech constants
- Greater for electrons
  - So nMOS gates leak more

- ❑ Negligible for older processes ( $t_{\text{ox}} > 20 \text{ \AA}$ )
- ❑ Critically important at 65 nm and below ( $t_{\text{ox}} \approx 10.5 \text{ \AA}$ )



# Junction Leakage

- ❑ Reverse-biased p-n junctions have some leakage
  - Ordinary diode leakage



# Diode Leakage

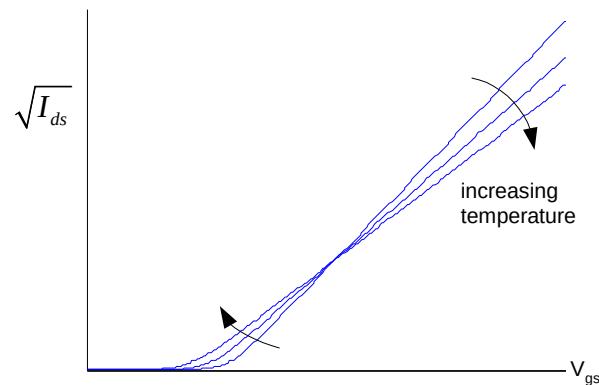
- ❑ Reverse-biased p-n junctions have some leakage

$$I_D = I_S \left( e^{\frac{V_D}{V_T}} - 1 \right)$$

- ❑ At any significant negative diode voltage,  $I_D = -I_S$
- ❑  $I_S$  depends on doping levels
  - And area and perimeter of diffusion regions
  - Typically  $< 1 \text{ fA}/\mu\text{m}^2$  (negligible)

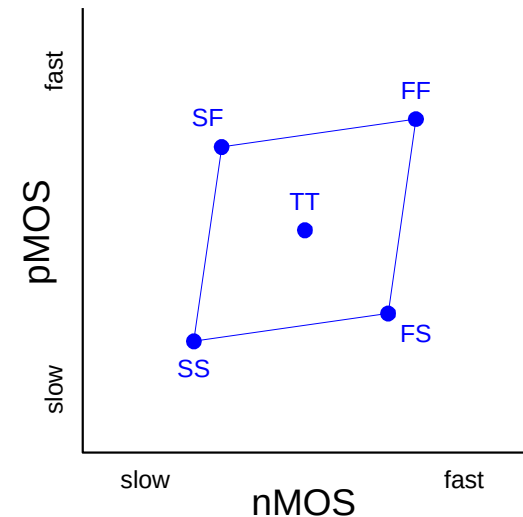
# Temperature Sensitivity

- ❑ Increasing temperature
  - Reduces mobility
  - Reduces  $V_t$
- ❑  $I_{ON}$  \_\_\_\_\_ with temperature
- ❑  $I_{OFF}$  \_\_\_\_\_ with temperature



# Parameter Variation

- ❑ Transistors have uncertainty in parameters
  - Process:  $L_{\text{eff}}$ ,  $V_t$ ,  $t_{\text{ox}}$  of nMOS and pMOS
  - Vary around typical (T) values
- ❑ Fast (F)
  - $L_{\text{eff}}$ : \_\_\_\_\_
  - $V_t$ : \_\_\_\_\_
  - $t_{\text{ox}}$ : \_\_\_\_\_
- ❑ Slow (S): opposite
- ❑ Not all parameters are independent for nMOS and pMOS



# Environmental Variation

- ☐  $V_{DD}$  and  $T$  also vary in time and space
- ☐ Fast:
  - $V_{DD}$ : \_\_\_\_\_
  - $T$ : \_\_\_\_\_

| Corner | Voltage | Temperature |
|--------|---------|-------------|
| F      |         |             |
| T      | 1.8     | 70 C        |
| S      |         |             |



# Process Corners

- ❑ Process corners describe worst case variations
  - If a design works in all corners, it will probably work for any variation.
- ❑ Describe corner with four letters (T, F, S)
  - nMOS speed
  - pMOS speed
  - Voltage
  - Temperature

# Important Corners

- ❑ Some critical simulation corners include

| Purpose              | nMOS | pMOS | $V_{DD}$ | Temp |
|----------------------|------|------|----------|------|
| Cycle time           |      |      |          |      |
| Power                |      |      |          |      |
| Subthreshold leakage |      |      |          |      |